AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19 /Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.

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PART 1: Multiple Choice Questions [10 Marks]

Questions 1 [10 marks] (1 mark/each):

1. **What does NIST SP 80-122 provide guidance on?**
   A) Encryption methods
   B) Privacy impact assessment
   C) Protecting the confidentiality of personally identifiable information
   D) Cloud computing security

2. **What is the purpose of a cryptographic hash function?**
   A) To encrypt data
   B) To generate a fixed-size string from input data
   C) To decrypt messages
   D) To create digital signatures

3. **Which of the following is a primary objective of system access controls?**
   a) To maximize system performance
   b) To ensure authorized users can access the system
   c) To enhance user interface design
   d) To minimize data storage costs

4. **Which of the following is an example of personally identifiable information (PII)?**
   A) Birthdate
   B) IP address
   C) Social Security number
   D) All of the above

5. **What is the purpose of a privacy risk assessment?**
   A) To determine system usability
   B) To identify the cost of system development
   C) To estimate the potential cost to the organization of privacy violations
   D) To increase system performance

6. **What is an example of an involuntary mechanism for obtaining PII?**
   a) Loyalty cards
   b) Social media posts
   c) Cookies on web browsers
   d) Voluntary surveys

7. **A Certificate Authority (CA) in PKI is responsible for:**
   A) Encrypting data with a public key
   B) Issuing digital certificates
   C) Providing symmetric keys for encryption
   D) Decrypting data with a private key

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8. **Privacy by Design involves:**
   A) Integrating privacy from the start of product development
   B) Adding privacy features as a final step
   C) Ignoring privacy concerns entirely
   D) Outsourcing privacy concerns to third parties

9. **Which type of access control ensures that users have the minimum level of access necessary to perform their job functions?**
   A) Role-based access control
   B) Mandatory access control
   C) Discretionary access control
   D) Least privilege access control

10. **Which of the following is a common method to secure data in transit?**
    A) Encryption
    B) Physical locks
    C) Visual surveillance
    D) Verbal agreements

PART 2: Short Answer Questions [20 Marks]

P2-Q1 [5 marks]: What is the main goal of a privacy risk assessment?

P2-Q2 [5 marks]: Define "Privacy by Design" (PbD).

P2-Q3 [5 marks]: Explain how cloud infrastructure vulnerabilities can lead to data breaches.

P2-Q4 [5 marks]: Explain the concept of de-anonymization.

PART 3: Problem Solving / Critical Thinking Questions [30 Marks]

P3-Q1 [9 marks]: You are appointed as the security manager at the Arab Open University. Based on this role, answer the following questions:

P3-Q1-1 [3 marks]: If your team requested your decision to decide between using symmetric or asymmetric encryption for securing data transmission. What factors would you consider in making this decision, and which type would you choose? Explain your reasoning.

P3-Q1-2 [6 marks]: You discovered that an employee has access rights that exceed their job requirements. How would you address this issue to align with the principle of least privilege?

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P3-Q2 [10 marks]: Given the following case-study on Autonomous Vehicles:

"Imagine a scenario where autonomous vehicles (self-driving cars) are widely used in Riyadh city. These vehicles are capable of navigating through traffic, following speed limits, and avoiding obstacles without human intervention. They rely on a combination of sensors, cameras, and artificial intelligence to make real-time decisions. Autonomous vehicles can also communicate with each other and with traffic infrastructure to optimize traffic flow and enhance safety. The vehicles collect and process vast amounts of data about their surroundings and the behavior of other road users."

P3-Q2-1 [5 marks]: How can autonomous vehicles be designed to ensure they only collect data that is essential for their operation and maintain user privacy?

P3-Q2-2 [5 marks]: 3. What security measures are essential to protect autonomous vehicles from being hacked or tampered with, ensuring the safety of passengers and other road users?

P3-Q3 [11 marks]: Given the following scenario, construct an Access Control Matrix for a file system with three users (Alice, Bob, and Carol) and three files (File1, File2, and File3). The permissions are as follows:

- **Alice** has read and write access to **File1** and **File2** but no access to **File3**.
- **Bob** has read access to **File1**, write access to **File3**, and no access to **File2**.
- **Carol** has read access to **File2** and **File3** but no access to **File1**.

Draw an Access Control Matrix with the appropriate permissions (**R** for read, **W** for write, and "**-**" for no access).

End of questions

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